Glucagon

Mechanism of action

- Glucagon increases the blood glucose level by stimulating glycogenolysis (the breakdown of glycogen into glucose), predominantly within the liver.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Hypoglycaemia when the patient cannot safely swallow glucose/food and IV access cannot be obtained.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 1 mg IM once for an adult or child aged greater than or equal to five years.
- 0.5 mg IM once for a child aged less than five years.

Administration

- Dissolve the powder using the syringe within the kit and administer IM.
- The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

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Common adverse effects

- None.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 15-60 minutes.

Usual preparation

- Ampoule containing 1 mg as powder.

Pharmacokinetics

- Glucagon is predominantly excreted unchanged into bile and urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

- None.

Additional information

- Glucagon relies on stored glycogen being available to exert its effect and if glycogen stores are not available, glucagon may be ineffective. Examples include if the patient:
  - Does not have diabetes.
  - Has sepsis.
  - Is a young child.
  - Has undergone strenuous exercise.
  - Has not eaten food for more than 12 hours.
  - Is suffering from adrenal insufficiency.

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- Is suffering from chronic hypoglycaemia.
- Is suffering from alcohol-induced hypoglycaemia.

- Following glucagon administration the patient's glycogen stores will be depleted. For this reason it is important the patient eats food as described in the **'hypoglycaemia'** guideline.

- Glucagon also reduces the tone and motility of the smooth muscle in the gastrointestinal tract, but is not recommended in the setting of oesophageal obstruction.

- Glucagon is sometimes suggested as part of the treatment for bradycardia caused by beta-blockers because it stimulates cardiac cells via a mechanism that is independent of the beta receptor. However, glucagon has almost no role in the out-of-hospital setting because it rarely provides a sustained heart rate rise in addition to adrenaline and requires much higher doses than that carried by ambulance personnel.